

Lab 02 - Runtimes (Part 1)

Instruction:

- This lab is divided into two parts. The objective of this part (Part 1) is to:
 - Define the selection sort algorithm.
 - Verify the function is accurate.
 - Generate a runtime table (as a list) for the algorithm and derive its runtime function from the table.
- The selection sort algorithm is:
`SelectionSort(A)`
 01. for $i = 0$ to $A.length$
 01. $m \leftarrow i$
 02. assign m the index of the element with the minimum value in $A[i:A.length-1]$
 03. if $i \neq m$, then swap the elements with indices i and m
- Submit a single .cpp source file.
- The file may include only these libraries: iostream, string, vector — no others.
- Submit the file to your GitHub repository, inside a directory named Lab02.
- Cheating of any kind is prohibited and will not be tolerated.
- **Violating and/or failing to follow any of the rules will result in an automatic zero (0) for the lab.**

At the very beginning of your submission, add a comment containing your name and the date, to confirm you have read and understood the instructions above.

Grading

Task	Maximum Points	Points Earned
1	1.5	
2	0.5	
3	0.5	
Total	2.5	

Task 1

Define the function:

```
void SelectionSort(vector<int>& v)
```

that sorts *v* in ascending order, in place, using the selection sort algorithm shown above.

Task 2

Directly below your definition of `SelectionSort()`, add a comment block that generate the runtime table of `SelectionSort()` and derive its runtime function as we did have Note01.

Task 3

In the main function verify that your `SelectionSort()` function is accurate by:

1. Initializing a vector object of at least 5 elements with randomly generated, unordered integer values.
2. Invoking `SelectionSort()` with the vector object as its argument.
3. Checking whether the resulting vector's elements are in ascending order. Display the string **"successful"** only if the vector object is sorted correctly.